

fsoft2026 1DD 1: Iteration 2 – Technical Report

Domain Model, Glossary, Functional and Non-Functional Requirements, System Sequence Diagram

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Abstract. This technical report details the second iteration of the Med-Manager project, a local software system designed to manage vaccination center operations. It presents the system's foundational architecture, including the Domain Model and a comprehensive business glossary. Furthermore, it outlines the structural and behavioral design through class diagrams, use case specifications, and sequence diagrams.

Keywords: Business, C++, Vaccination Center, MedManager, Management, Software Development, Command-line Interface, Health

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1 Introduction

The **MedManager** project is a management system designed for a stand-alone local vaccination center.

In this second iteration, we advanced the system design by refining the domain model and establishing the core structural and behavioral diagrams. To support this, a comprehensive business glossary was defined. Furthermore, class diagrams were developed for the Model, View, Controller, and Exceptions components, alongside use case and sequence diagrams mapping system behavior across various user interactions.

This document details these components, the physical and logical architectures, and the overall code organization, reinforcing the system's structural foundation and validating our implementation decisions.

2 Domain Model

The Domain Model of the MedManager system defines and structures the fundamental concepts of the vaccination process. It translates the business requirements into a diagram that clearly shows the system's entities (such as users and nurses) and how they relate to each other, as shown in Figure 1.

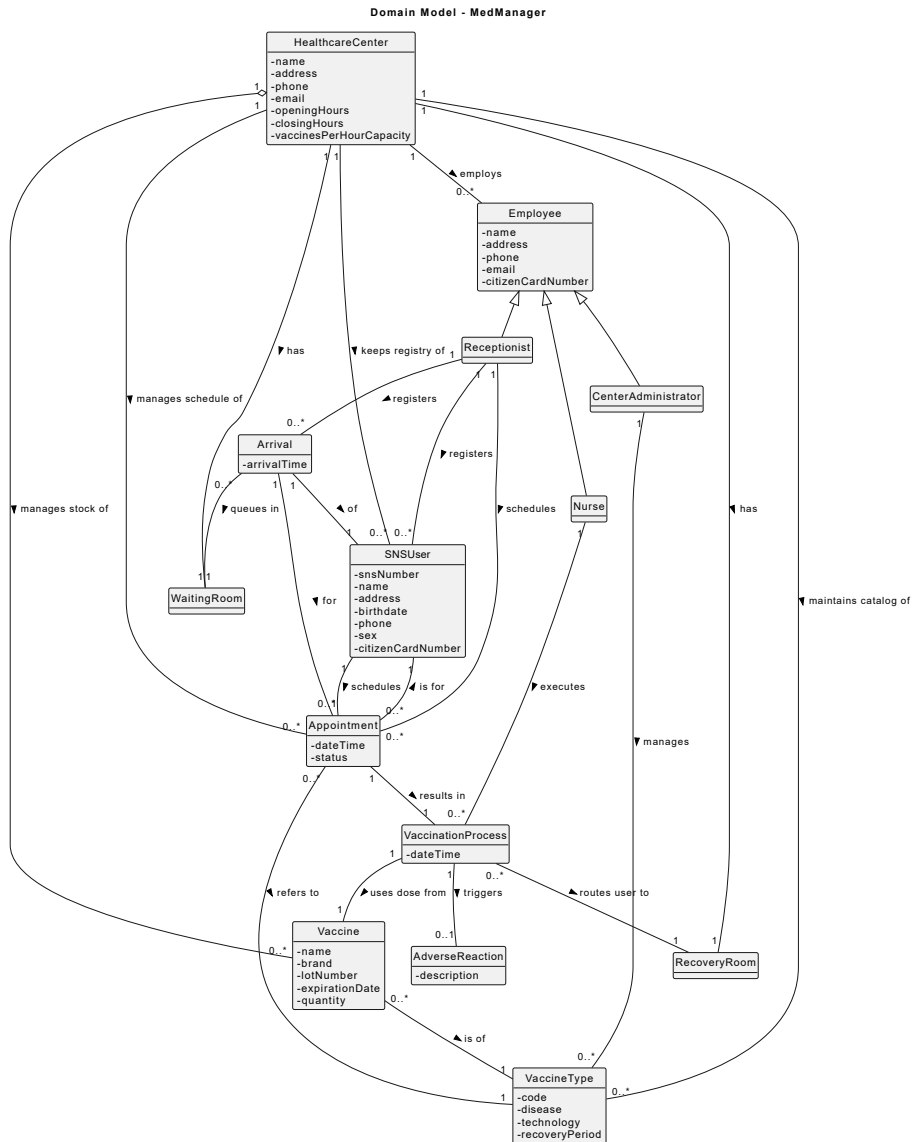


Fig. 1. MedManager’s Domain Model

3 Glossary

Table 1. MedManager Glossary

Term/Expression (EN)	Termo (PT)	Definition/Description
Adverse Reaction	Reação Adversa	Clinical event or side effect recorded after vaccination, typically monitored in the recovery room during the designated period.
Appointment	Agendamento	A scheduled slot (date and time) for an SNS User to receive a specific vaccine type. It serves as the prerequisite for a vaccination process.
Arrival	Chegada	The formal record of the user's physical presence at the center. It links an Appointment to the Waiting Room, triggering the service flow.
Center Administrator	Administrador do Centro	Employee responsible for managing center resources, including the vaccine catalog (types), physical stock, and employee records.
Employee	Funcionário	Any worker at the center (Administrator, Nurse, or Receptionist), identified by name, address, phone, email, and citizen card number.
Healthcare Center	Centro de Saúde	The facility where vaccinations occur, defined by its operating hours and its hourly vaccination capacity.
Nurse	Enfermeiro	Healthcare professional responsible for the clinical act of vaccination and for consulting the Waiting Room to call the next user.
Receptionist	Rececionista	Employee responsible for registering the arrival of users, confirming appointments, and managing initial user data.
Recovery Room	Sala de Recuperação	A monitored area where users must remain after vaccination for a period defined by the specific Vaccine Type to check for reactions.
SNS User	Utente do SNS	A citizen eligible for vaccination, identified by a unique SNS number, citizen card number, and birthdate.
Vaccination Process	Processo de Vacinação	The clinical act of administering a vaccine dose, linking a nurse, a specific vaccine lot, and an appointment.
Vaccine	Vacina	A physical unit in the center's inventory, characterized by a commercial name, brand, lot number, expiration date, and available quantity.
Vaccine Type	Tipo de Vacina	A category (e.g., COVID-19) that defines the technology used, the disease it prevents, and the required recovery period for users.
Waiting Room	Sala de Espera	A logical queue that manages arrivals in a First-Come, First-Served (FCFS) order until a nurse is available to perform the vaccination.

4 Design

4.1 Physical architecture

The system's physical architecture follows a localized deployment model, where the entire application is self-contained. As illustrated in Figure 2, the MedManager software component runs entirely on a single target node, represented by a local PC.

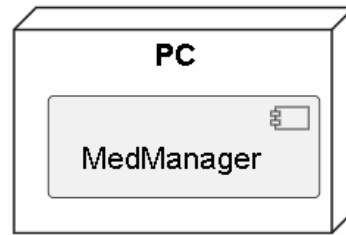


Fig. 2. Physical architecture

4.2 Logical architecture

The system's logical structure follows the MVC pattern within the MedManager package. The Controller serves as the central coordinator, linking the Model and View components through defined interfaces to ensure a clear separation of concerns, as shown in Figure 3.

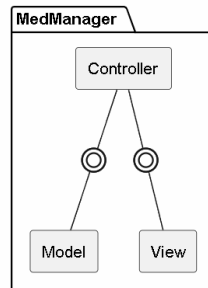


Fig. 3. Logical architecture

4.3 Code organization

Figure 4 illustrates the high-level architecture of MedManager, mapping the package dependencies between the controllers, views, model, and exceptions layers to guarantee a clean and decoupled codebase.

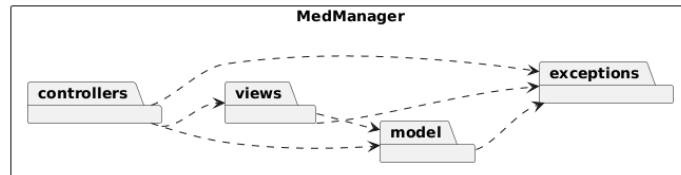


Fig. 4. Code Organization

4.4 Model - Class Diagram

Figure 5 maps out the core structural entities of the healthcare center system, detailing their attributes, operations, and relationships to define how clinical data and domain logic are stored and managed.

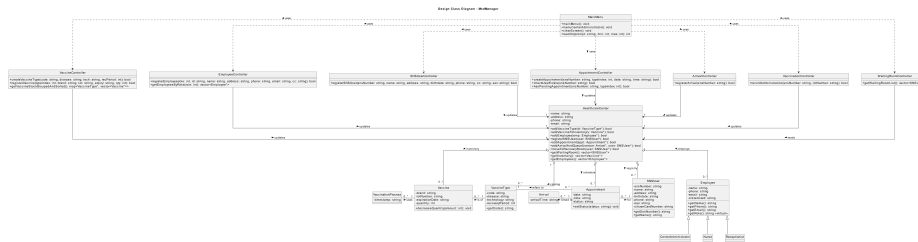


Fig. 5. Model - Class Diagram

4.5 Exceptions - Class Diagram

Figure 6 defines the system’s specialized error-handling strategy, detailing how custom exception classes inherit from `std::exception` to catch and manage invalid data or duplicate entries gracefully.

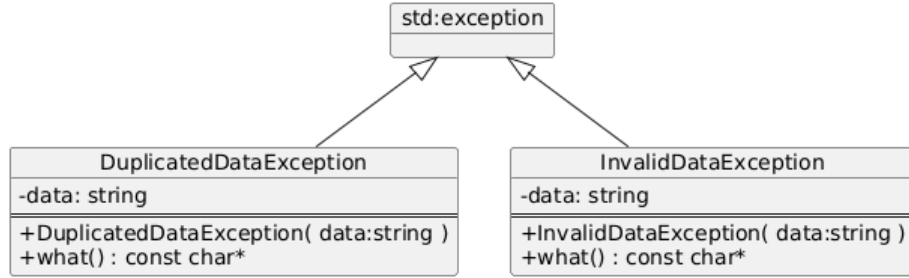


Fig. 6. Exceptions - Class Diagram

4.6 Views - Class Diagram

The **Views** layer of MedManager follows a centralized hierarchy where a main view delegates terminal input and output to specialized sub-views for each clinical context (vaccines, staff, patients, and schedules). Supported by static utility functions and a dedicated input exception handler, this layer completely isolates the user interface, safely capturing data before passing it to the controller, as illustrated in Figure 7.

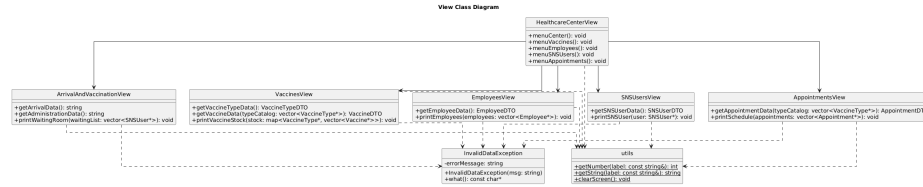


Fig. 7. Views - Class Diagram

4.7 Controllers - Class Diagram

The **Controller** acts as the single orchestrator of the system, centralizing the execution flow by connecting the `HealthcareCenter` model with the presentation layer. It manages user navigation and maps menu inputs directly to specific internal sub-routines like staff registration, scheduling, or vaccine management. Details are shown in Figure 8.

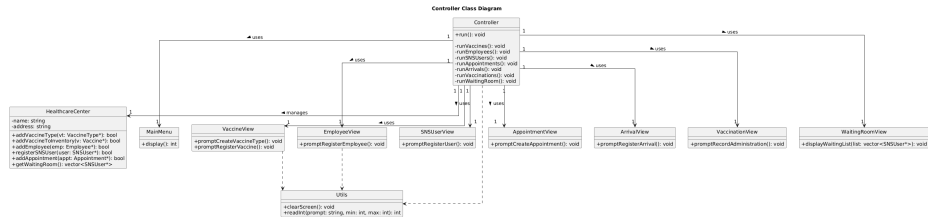


Fig. 8. Controllers - Class Diagram

4.8 User Interface

Figure 9 represents the general structure of the menus and their interactions within the system.

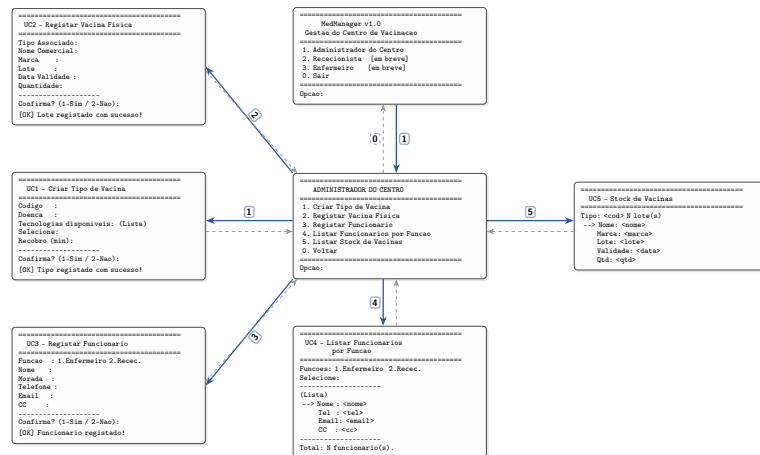


Fig. 9. User Interface Design

4.9 Use Case Diagrams

The system's functional requirements are driven by four primary actors, each with distinct roles and permissions within the healthcare center. The following diagrams illustrate the specific use cases available to the Center Administrator, Receptionist, SNS User, and Nurse.

As shown in Figure 10, the **Center Administrator** is responsible for the core management of the system, including managing the vaccine catalog, stock inventory, and registering staff members.

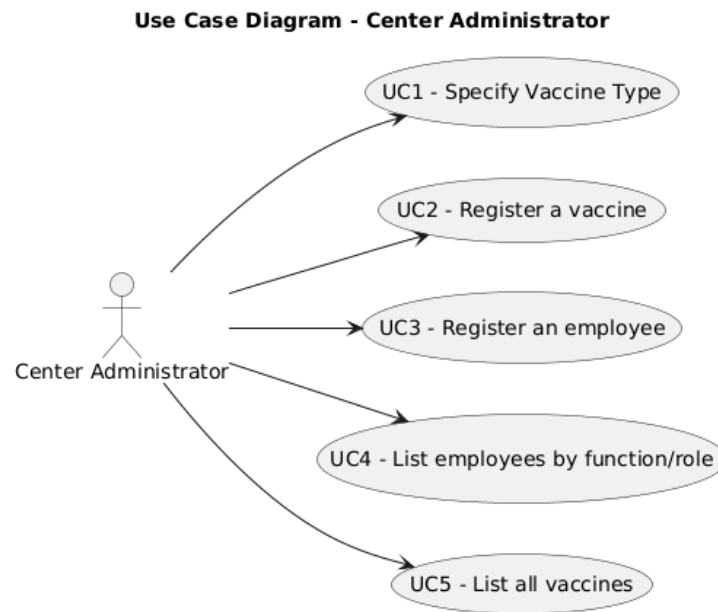


Fig. 10. Use Case Diagram - Center Administrator

Figure 11 depicts the **Receptionist's** interactions, which focus on front-desk operations such as registering new SNS Users, scheduling appointments on their behalf, and registering their physical arrival at the center.

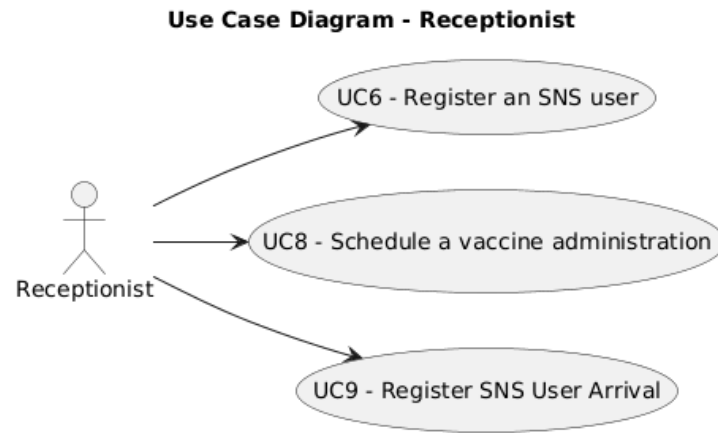


Fig. 11. Use Case Diagram - Receptionist

The **SNS User** has a more restricted set of interactions, primarily dealing with self-scheduling their own vaccination appointments, as illustrated in Figure 12.

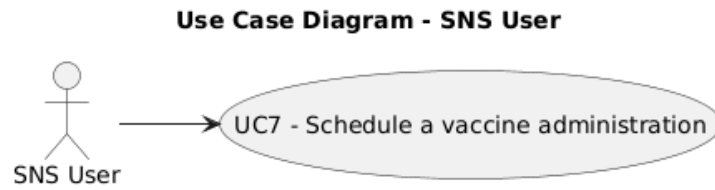


Fig. 12. Use Case Diagram - SNS User

Finally, Figure 13 details the **Nurse's** workflow, which involves consulting the waiting room queue and recording the actual clinical administration of vaccines to patients.

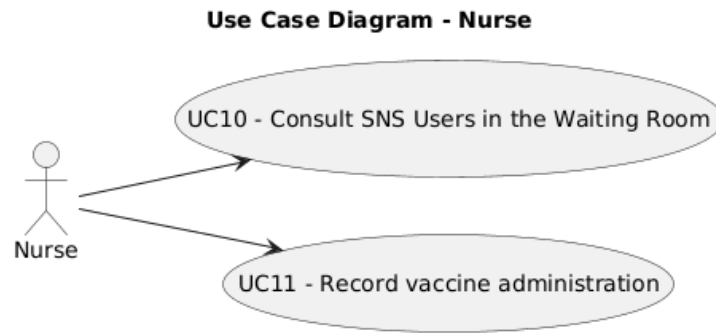


Fig. 13. Use Case Diagram - Nurse

4.10 Sequence Diagram

Use Case 1: Create Vaccine Type

The specifications for this use case are detailed in Table 4.10, and its execution flow is visually represented in the sequence diagram in Figure 14.

tableUse Case 1: Create Vaccine Type

Actor	Center Administrator
Use case name	Create Vaccine Type
Description	The Center Administrator specifies a new vaccine type to be added to the healthcare center's catalog.
Precondition	The user must be logged in as Center Administrator.
Post-condition	A new Vaccine Type is created and added to the vaccine catalog.
Main flow	1. User starts specifying a new vaccine type. 2. System requests data (code, disease, recoveryPeriod) and presents a list of technologies. 3. User inputs data (code, disease, recoveryPeriod) and selects a technology. 4. System validates the code uniqueness, registers the new vaccine type in the catalog, and confirms successful creation.
Alternative path	If the code validation fails (i.e., the vaccine code already exists), the system informs the user and resumes at step 2 (or aborts the operation).

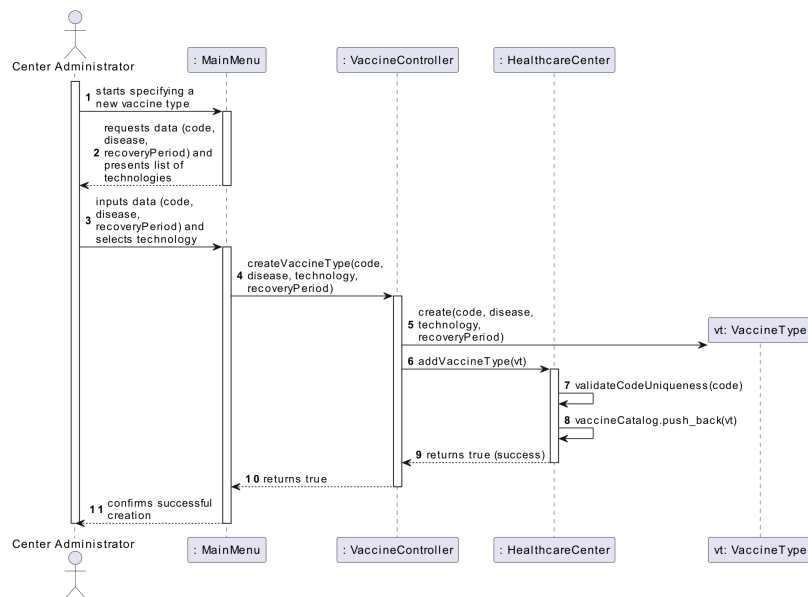


Fig. 14. UC1: Create Vaccine Type

Use Case 2: Register Vaccine

The specifications for this use case are detailed in Table 4.10, and its execution flow is represented in Figure 15.

tableUse Case 2: Register Vaccine

Actor	Center Administrator
Use case name	Register Vaccine
Description	The Center Administrator registers a batch of specific vaccines into the center's inventory, linking it to an existing vaccine type.
Precondition	The user must be logged in as Center Administrator and the corresponding Vaccine Type must already be registered.
Post-condition	A new batch of vaccines is registered and added to the center's inventory.
Main flow	1. User starts the vaccine registration. 2. System fetches the vaccine catalog and shows a list of available Vaccine Types. 3. User selects the desired vaccine type and inputs the batch data (brand, lot, expiry, quantity). 4. System creates the vaccine batch, adds it to the inventory, and confirms successful registration.
Alternative path	If any inputted data format is invalid or required fields are missing, the system informs the user and resumes at step 2.

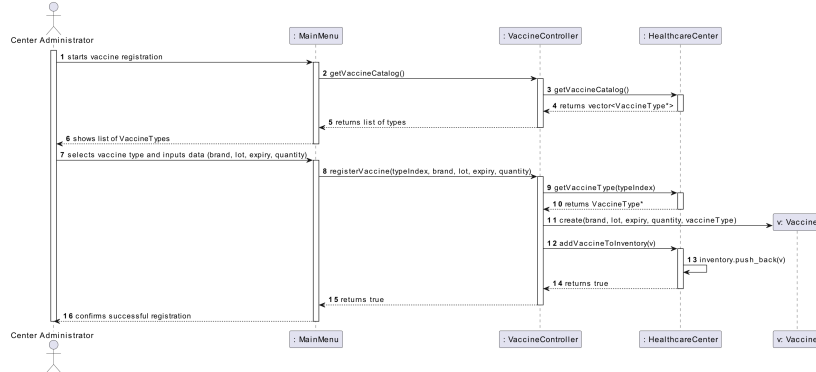


Fig. 15. UC2: Register Physical Vaccine

Use Case 3: Register New Employee

The specifications for this use case are detailed in Table 4.10, and its execution flow is represented in Figure 16.

tableUse Case 3: Register New Employee	
Actor	Center Administrator
Use case name	Register New Employee
Description	The Center Administrator registers a new center staff member, specifying their institutional role (Nurse or Receptionist).
Precondition	The user must be logged in as Center Administrator.
Post-condition	A new Employee profile (Nurse or Receptionist) is registered and added to the center's staff records.
Main flow	1. User starts the registration of a new employee. 2. System shows the available roles (Nurse, Receptionist) and asks to select one. 3. User selects the intended role. 4. System requests personal data (id, name, address, phone, email, cc). 5. User inputs the requested data. 6. System checks for duplicates and creates the employee entity according to the selected role. 7. System adds the new employee to the records and confirms successful registration.
Alternative path	If the system detects duplicated data (e.g., phone, email, or cc already exists), the registration is aborted and an error is displayed.

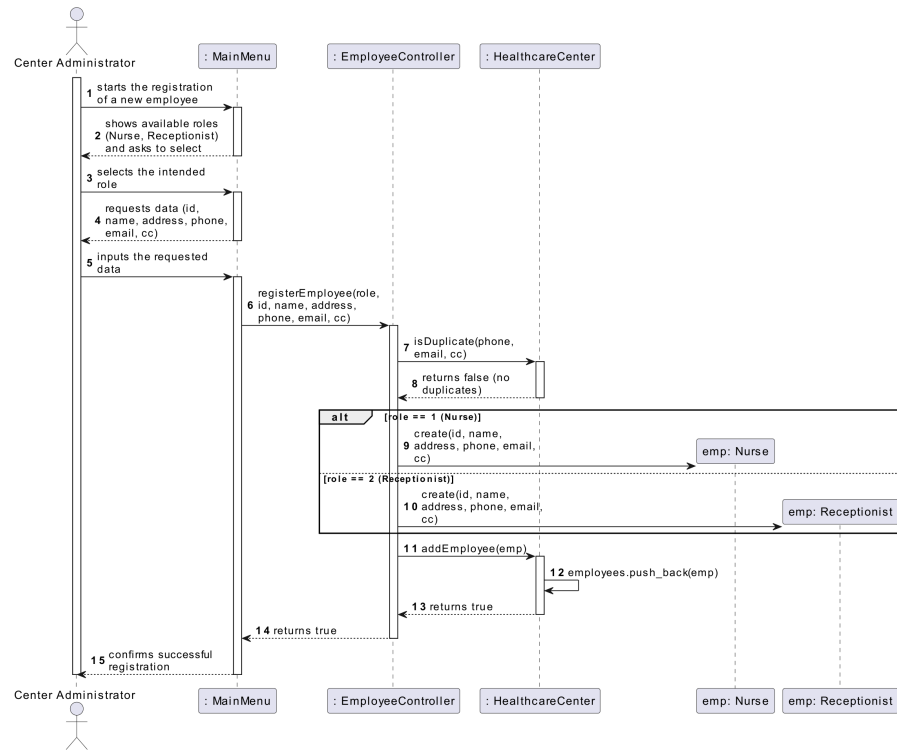


Fig. 16. UC3: Register Employee

Use Case 4: List Employees by Role

The specifications for this use case are detailed in Table 4.10, and its execution flow is represented in Figure 17.

tableUse Case 4: List Employees by Role

Actor	Center Administrator
Use case name	List Employees by Role
Description	The Center Administrator requests and views a list of registered employees filtered by a specific role (Nurse, Receptionist, or Admin).
Precondition	The user must be logged in as Center Administrator.
Post-condition	The system presents the filtered list of employees or informs that no matches were found.
Main flow	1. User requests a list of employees. 2. System prompts the user to select a role (Nurse, Receptionist, Admin). 3. User selects the desired role. 4. System fetches all employees, filters them by the selected role through an internal loop, and returns a filtered list. 5. System displays the filtered list showing each employee's details (Name, Phone, E-mail, CC).
Alternative path	If the filtered list is empty, the system informs the user that no employees are registered with that specific role.

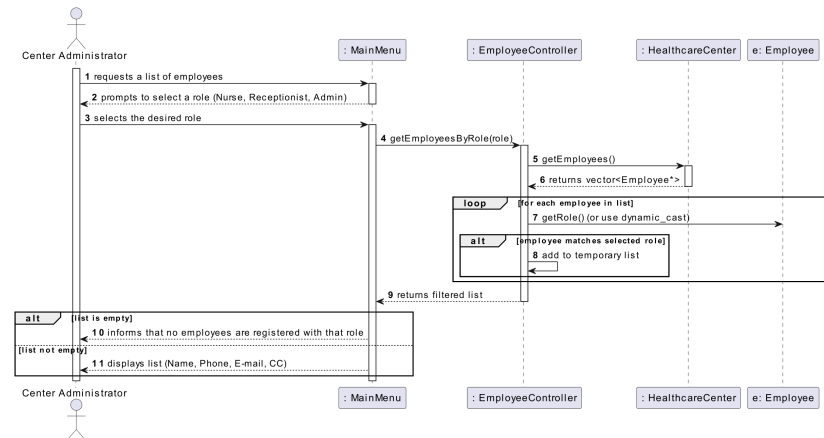


Fig. 17. UC4: List Employees by Role

Use Case 5: List Vaccine Stock Grouped and Sorted

The specifications for this use case are detailed in Table 4.10, and its execution flow is represented in Figure 18.

tableUse Case 5: List Vaccine Stock Grouped and Sorted

Actor	Center Administrator
Use case name	List Vaccine Stock Grouped and Sorted
Description	The Center Administrator requests and views a list of all vaccines in stock, grouped by vaccine type and sorted alphabetically by brand.
Precondition	The user must be logged in as Center Administrator.
Post-condition	The system presents the structured list of vaccines or informs that the inventory is empty.
Main flow	1. User requests a list of all vaccines. 2. System fetches the inventory and checks if it contains items. 3. System groups the vaccines by VaccineType and sorts each group alphabetically by Brand. 4. System displays the vaccines grouped by type, including the Commercial Name, Brand, Lot Number, Expiration Date, and Quantity for each.
Alternative path	If the inventory is empty, the system returns an empty structure and displays a "No inventory found" message.

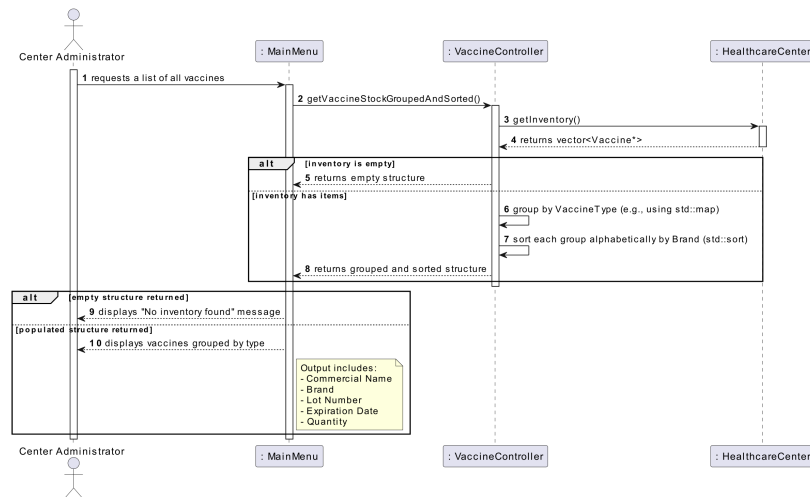


Fig. 18. UC5: List Vaccines Stock

Use Case 6: Register New SNS User

The specifications for this use case are detailed in Table 4.10, and its execution flow is represented in Figure 19.

tableUse Case 6: Register New SNS User

Actor	Receptionist
Use case name	Register New SNS User
Description	The Receptionist registers a new National Health Service (SNS) user profile in the system's database.
Precondition	The user must be logged in as Receptionist.
Post-condition	A new SNS User profile is created and successfully stored in the health center registry.
Main flow	1. User starts the registration of a new SNS user. 2. System requests personal data (Name, Address, Phone, SNS Number, CC, Birth Date, Sex [Optional]). 3. User inputs the requested data. 4. System checks for user duplication by SNS number and phone, and validates the Portuguese contact/identity formats. 5. System creates the SNS user entity, appends it to the registry records, and confirms successful registration.
Alternative path	If the system detects an invalid format or an existing duplicate, it returns a validation error and informs the user.

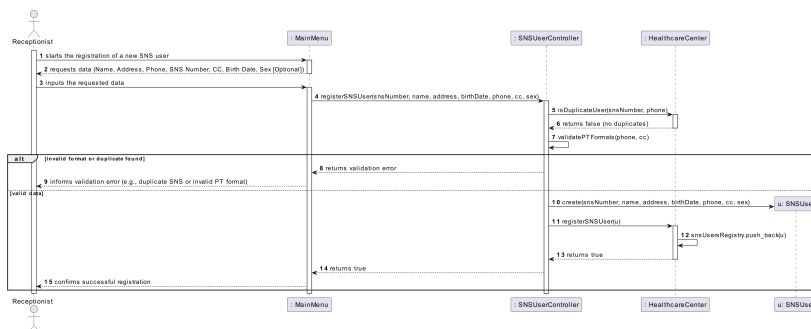


Fig. 19. UC6: Register an SNS User

Use Case 7: Schedule Vaccine Administration

The specifications for this use case are detailed in Table 4.10, and its execution flow is represented in Figure 20.

tableUse Case 7: Schedule Vaccine Administration

Actor	SNS User
Use case name	Schedule Vaccine Administration
Description	The SNS User schedules a vaccine administration by verifying their registration, choosing a vaccine type, and selecting an available date and time.
Precondition	The user must be registered in the system as a valid SNS User.
Post-condition	A new appointment is successfully scheduled and stored in the healthcare center's master schedule.
Main flow	1. User starts scheduling a vaccine administration. 2. System requests the SNS User Number. 3. User inputs the SNS User Number. 4. System verifies that the user exists, retrieves the vaccine catalog, and presents a list of available Vaccine Types. 5. User selects a vaccine type and inputs the desired Date and Time. 6. System validates the selected date/time, displays an appointment summary, and requests confirmation. 7. User confirms the appointment. 8. System links the appointment details, creates the appointment record, appends it to the master schedule, and confirms successful scheduling.
Alternative path	- If the user is not registered, the system informs a user not found error and aborts. - If the date/time is in the past, the system informs an invalid date/time error.

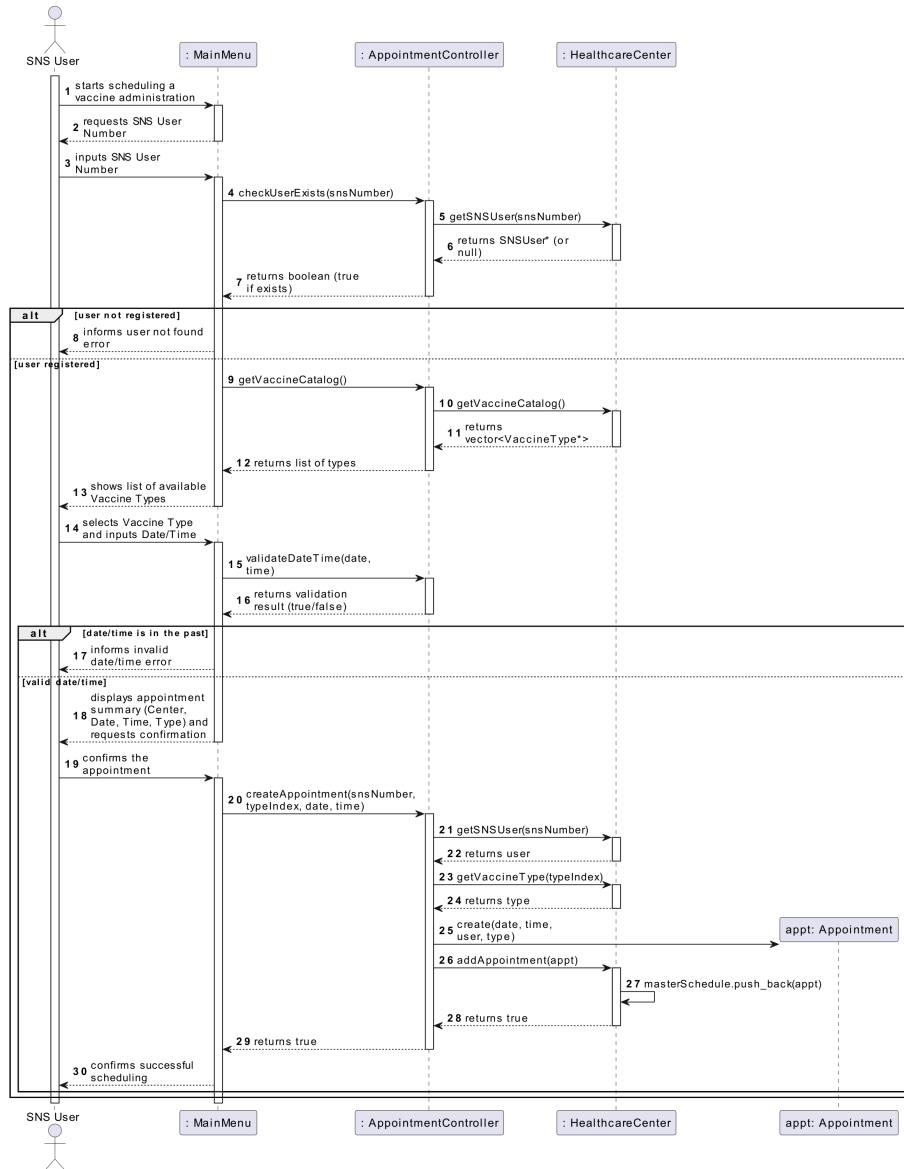


Fig. 20. UC7: Schedule a Vaccine Administration (SNS User)

Use Case 8: Schedule Vaccine Administration for an SNS User

The specifications for this use case are detailed in Table 4.10, and its execution flow is represented in Figure 21.

tableUse Case 8: Schedule Vaccine Administration for an SNS User

Actor	Receptionist
Use case name	Schedule Vaccine Administration for an SNS User
Description	The Receptionist schedules a vaccine administration on behalf of an SNS User, verifying their identity and ensuring no pending appointments conflict with the selection.
Precondition	The user must be logged in as Receptionist and the target SNS User must exist in the system database.
Post-condition	A new appointment is created and registered in the healthcare center's master schedule.
Main flow	1. User starts scheduling a vaccine administration for an SNS User. 2. System requests the SNS User Number. 3. User inputs the requested SNS User Number. 4. System validates the user's existence, retrieves the available Vaccine Types, and displays them. 5. User selects the intended Vaccine Type and inputs the Date and Time. 6. System checks for scheduling conflicts or pending appointments, displays an appointment summary, and asks for confirmation. 7. User confirms the appointment parameters. 8. System instantiates the new appointment record, stores it in the master schedule, and confirms successful scheduling.
Alternative path	- If the target SNS User is not registered, the system returns a user not found error. - If the date/time is invalid or a pending appointment already exists for that user and vaccine type, the system displays a scheduling error.

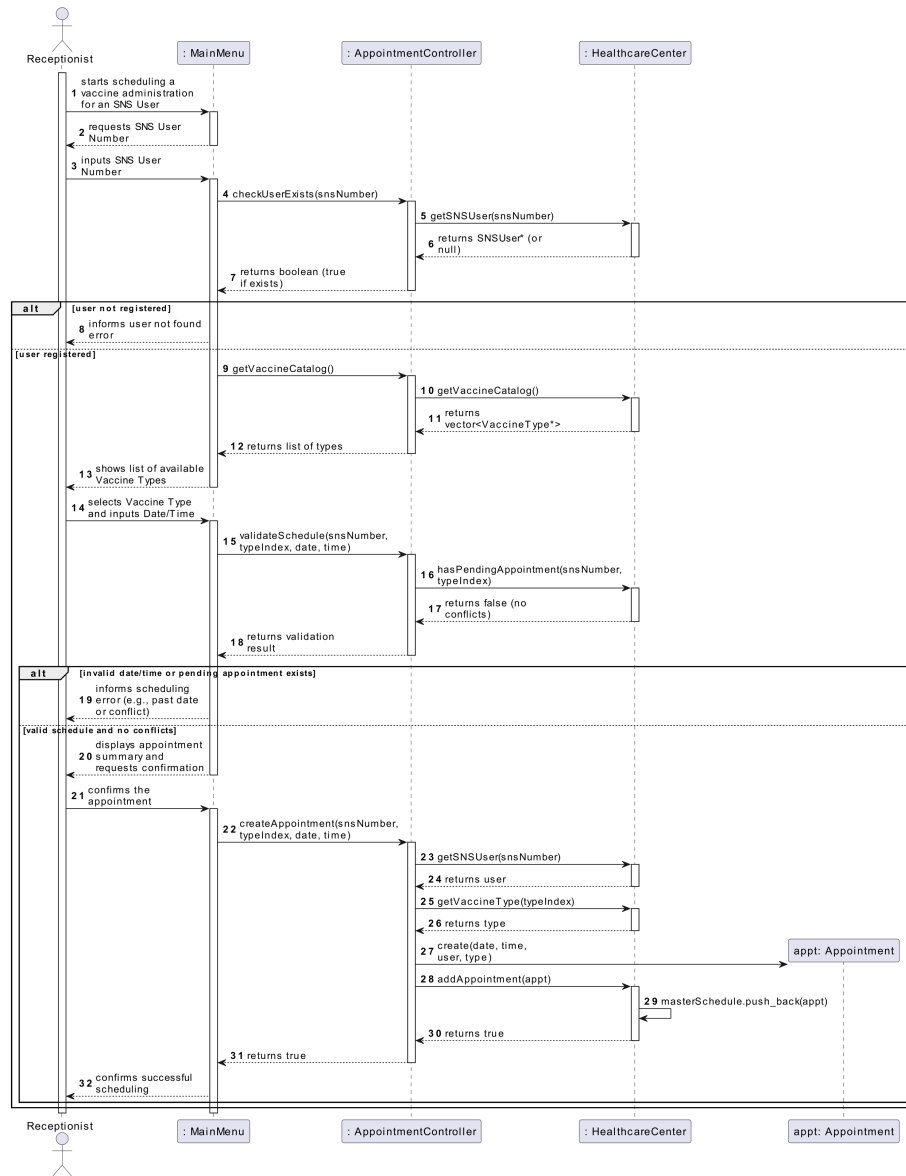


Fig. 21. UC8: Schedule a Vaccine Administration (Receptionist)

Use Case 9: Register Arrival of an SNS User

The specifications for this use case are detailed in Table 4.10, and its execution flow is represented in Figure 22.

tableUse Case 9: Register Arrival of an SNS User

Actor	Receptionist
Use case name	Register Arrival of an SNS User
Description	The Receptionist registers the arrival of an SNS User at the healthcare center, confirming they have an appointment for the current day and are not already in the waiting room.
Precondition	The user must be logged in as Receptionist.
Post-condition	The user's status is updated to waiting, an arrival record is created, and the user is added to the waiting room queue.
Main flow	1. User starts registering the arrival of an SNS user. 2. System requests the SNS User Number. 3. User inputs the requested SNS User Number. 4. System checks if there is an appointment scheduled for today and verifies that the user is not already in the waiting room. 5. System updates the appointment status to "WAITING". 6. System creates an arrival record with the current time. 7. System adds the arrival to the center's registry, moves the user into the waiting room queue, and confirms successful registration and assignment.
Alternative path	If the user has no appointment scheduled for today or is already registered in the waiting room, the system returns a validation error and informs the user.

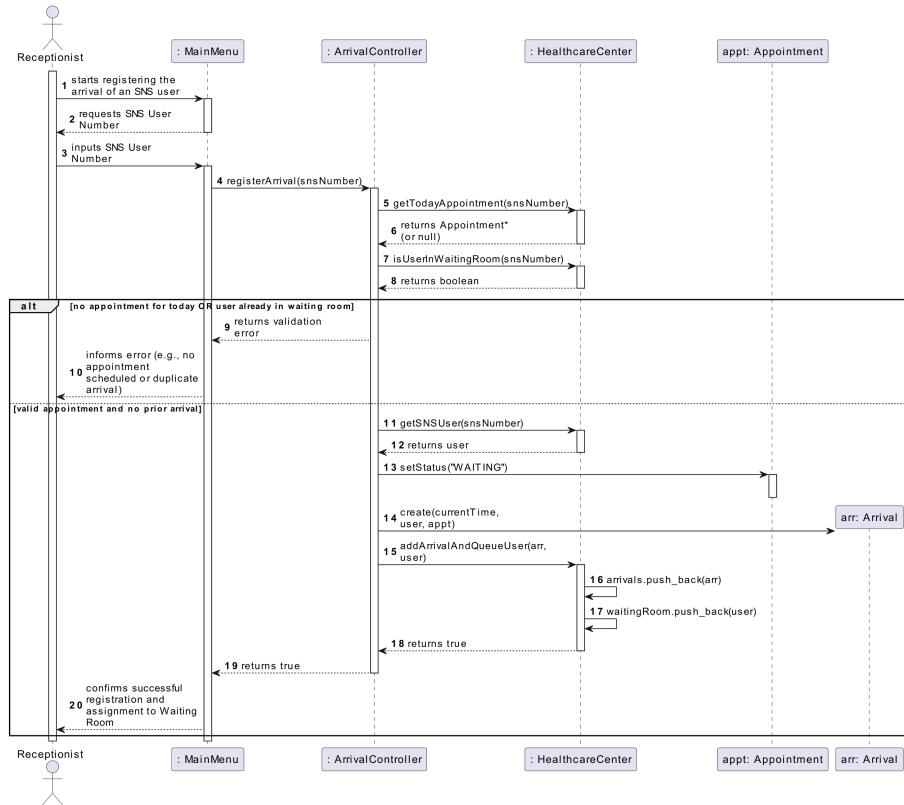


Fig. 22. UC9: Register Arrival of an SNS User

Use Case 10: List Users in the Waiting Room

The specifications for this use case are detailed in Table 4.10, and its execution flow is represented in Figure 23.

tableUse Case 10: List Users in the Waiting Room

Actor	Nurse
Use case name	List Users in the Waiting Room
Description	The Nurse requests and views the list of SNS Users currently inside the healthcare center's waiting room, displayed in first-in, first-out (FIFO) order.
Precondition	The user must be logged in as Nurse.
Post-condition	The system displays the active queue of waiting users or informs that the waiting room is empty.
Main flow	1. User requests the list of users in the waiting room. 2. System fetches the current waiting room list from the healthcare center data records. 3. System processes each user in the list through a loop to retrieve their identification attributes (SNS Number and Name). 4. System displays the complete list of users in FIFO order, showing each individual's SNS Number and Name.
Alternative path	If the waiting room is empty, the system displays a "Waiting room is currently empty" message.

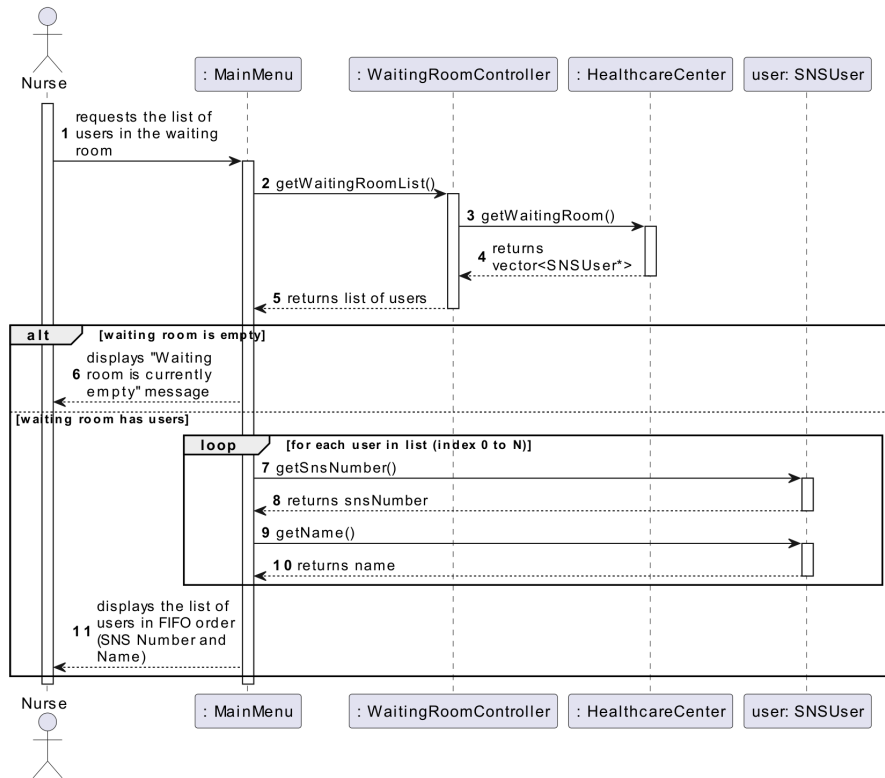


Fig. 23. UC10: Consult Waiting Room

Use Case 11: Record Vaccine Administration

The specifications for this use case are detailed in Table 4.10, and its execution flow is represented in Figure 24.

tableUse Case 11: Record Vaccine Administration

Actor	Nurse
Use case name	Record Vaccine Administration
Description	The Nurse verifies the presence of an SNS User in the waiting room, selects an available vaccine lot, and records the administration, automatically transitioning the user to the recovery room.
Precondition	The user must be logged in as Nurse.
Post-condition	The vaccine stock quantity is decremented, the appointment status is updated to administered, a vaccination process record is stored, and the user is transferred to the recovery room.
Main flow	1. User starts recording a vaccine administration. 2. System requests the SNS User Number. 3. User inputs the SNS User Number. 4. System checks if the user is currently in the waiting room, fetches the available vaccine stock inventory, and displays a list of Vaccines and Lot Numbers. 5. User selects the administered Vaccine and Lot Number. 6. System updates the user's appointment status to administered and decreases the vaccine stock quantity by 1. 7. System creates a new VaccinationProcess record with the current time and links it to the center's records. 8. System removes the user from the waiting room, adds them to the recovery room queue, and confirms successful administration and transfer.
Alternative path	If the system detects that the user is not in the waiting room, it displays an error message and terminates the process.

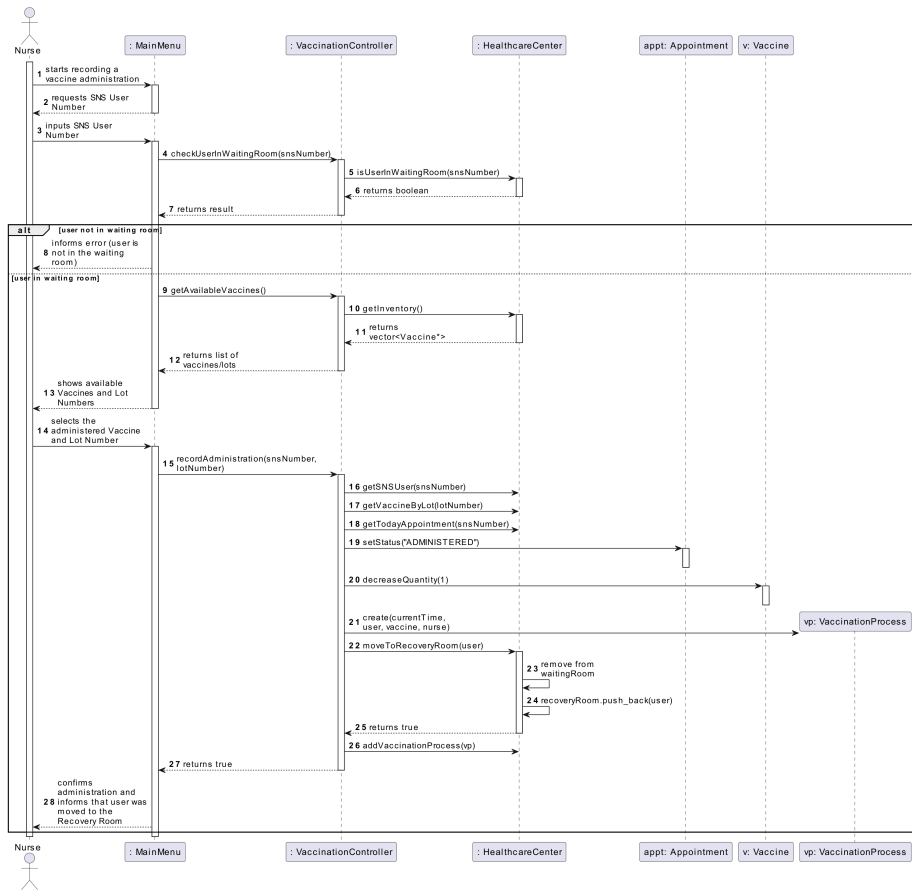


Fig. 24. UC11: Record Vaccine Administration

5 Conclusion

In conclusion, Iteration 2 of the **MedManager** project successfully established a robust architectural and structural foundation for the local vaccination system. By refining the Domain Model and detailing the system’s Logical Architecture through the MVC pattern, a clear separation of concerns was ensured across the Model, View, Controller, and Exceptions layers.

Furthermore, the comprehensive specification of the eleven core Use Cases—ranging from inventory and staff management to patient scheduling and vaccine administration—alongside their respective Sequence Diagrams, provides a precise blueprint of system behavior and user interactions.

These design artifacts not only validate the initial functional requirements but also mitigate future development risks by clearly defining data flows, validations, and object responsibilities. Moving forward, the structural guidelines and diagrams finalized in this iteration will serve as the primary baseline for the C++ implementation, ensuring that the resulting software is modular, maintainable, and fully aligned with the daily operational needs of the healthcare center.

6 References

1. P. B. Sousa, “Lecture Slides for FSOF (Fundamentals of Software Engineering),” Lecture notes and presentation slides, Bachelor in Telecommunications and Informatics Engineering, ISEP (School of Engineering, Polytechnic of Porto), 2025–2026. Provided by the instructor.